Adaptive Systems for Hallucination Detection of LLMs using Lie Group Methods

Kirill Nevzorov under supervision
of Prof. Christopher Buckley
University of Sussex
School of Engineering and Informatics

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Abstract

We wish to research the process of creating and use-case testing various LLM prompts, such as prompting the user for additional details relevant to their project, assisting them in resumarising and elaborating on the information stored within a workspace, and generating intelligence from a collection of notes.

Contents

Foreword

$$z_{\mathrm{cr}}^{\mathrm{Hz}} = \left(\sum_{i=1}^{N-1} \mathbf{1}\{s_{i-1}s_i < 0\} \right) \frac{f_s}{2(N-1)}$$

where $s = (s_0, \ldots, s_{N-1})$ is the discrete signal of length N and f_s is the sampling frequency.

In previous years of study, I have produced `pagerts`, an application with behaviour similar to a web scraper. I was curious how its build process affected the runtime of the application. When installing it to the system repository it would somehow begin to exhibit correct behaviour, yet running it from the local installation folder it would perform like a highly unoptimised code.

To address the issue and report on the matter, which was only possible to produce through trial and error, I had discovered that running the package from system shared library path was the correct approach.

Conversely, running from development folder it would have to call across multiple system boundaries without any optimisations in order to open the code necessary to function, rendering its runtime nondeterministic.

Introduction

A terminal display only displays the information that a developer has agreed to output to the terminal, as often as it has been specified in the sources, with as much accuracy as they last touched the code.

Effective software design often relies on iterative and incremental refinement of both code and text-based documentation. Software and various command line tools have been known to misehave since the beginning of time.

However, untraditional workflows can fragment the code and thought process the developers construct while writing an application, requiring constant context switching between development environments, notetaking applications, and external tools.

A novel way to take notes with the help of LLMs (Large Language Models) is proposed, and we wish to make the process of interacting with the AI as deterministic as possible. We aim to expand on existing techniques by implementing a process of procedural validation, parsing, and aim to create an application that is able to aid the user in a predictable manner, harnessing the power of text transformers.

In previous years of study, I have produced `pagerts`, an application with behaviour similar to a web scraper. I was curious how its build process affected the runtime of the application. When installing it to the system repository it would somehow begin to exhibit correct behaviour, yet running it from the local installation folder it would perform like a highly unoptimised code.

To address the issue and produce the report document elaborating on the matter, which was only possible to produce through trial and error, I had to discover that the package was running effectively from system shared library path, yet when installed as a `node.js` package in a development folder it would have to call across multiple system boundaries in order to open the code necessary to function effectively, rendering its runtime performance unpredictable.

Background

Layout and format of the notes paired with markup is often necessary in order to make an approach of looking into the information written down effective, yet no commercially-available process markets itself as primarily-oriented around the process of writing down the information in as much detail as possible, while minimising user error.

There exist various strategies for taking notes, ranging from disciplines such as Zettelkasten, which focus on the workflow, to Markdown, which focus on the format. In this report we will explore `Ollama`, an application ecosystem based around AI and LLMs ran on local hardware with the purpose of constructing a chat messaging app. The documentation leads the user to either implement one from scratch, or to discover one of the many frontend applications already developed.

Professional Considerations

As a professional body, the British Computer Society (known as BCS, the Chartered Institute for IT), has a responsibility to set rules and professional standards to direct the behaviour of its members in professional matters. It is expected that these rules and professional standards will be higher than those established by the general law and that they will be enforced through disciplinary action which can result in expulsion from membership.

Members are expected to exercise their own judgement (which should be made in such a way as to be reasonably justified) to meet the requirements of the code and seek advice if in doubt.

Appendix 1 to the code sets examples of interpretation of the tenets of professional conduct and form part of the Code of Conduct.

See the original statement online.

Motivation

Our motivation for this project comes directly from the point of view that a creative block, writers paralysis, demotivation and low mood which can directly affect the cadence of any given project but can be tackled with the power of a carefully engineered software aimed at increasing productivity of the creator may be solved with a conversational agent, which can be instructed to coordinate the operator's activities.

This project aims to fill this gap by developing a suite of tools that can be integrated into a note-taking web-application, or to serve as a framework for specifying a toolkit in future work.

Literature Review

Lodde Networks

Network visualisation has been extensively studied for representing complex relationships between entities, with significant work focusing on web structures and hierarchies. Lodde (2009) provides a comprehensive analysis of various techniques including force-directed layouts and hierarchical edge bundling (Lodde 2009).

While these methods offer valuable insights into network structure, as web pages on the Internet are no longer static, a unique challenge presents itself, such that traditional network analysis insufficient.

The field has emphasised the importance of creating visualisations that support system administrators in their daily tasks. As Lodde notes: "Visualisation tools should support the user in fields of security handling, flow monitoring and the status of the system". This principle is particularly relevant to web analysis, where understanding resource relationships and dependencies is crucial for effective site navigation and content maintenance.

The growing importance of mobile access to web tools highlights the need for responsive visualisation access solutions. As noted by Lodde (2009), traditional desktop visualisation techniques face significant limitations in mobile contexts, particularly regarding screen real estate and touch-based interaction patterns.

To address this limitation, it is possible to develop software interfaces that avoid adhering to any prescriptive design principle, and instead use a combination of best practices, at all times.

Lie Group Theory

Critical Gap Analysis

Current literature demonstrates that while network visualisation techniques exist for various domains, there is limited research specifically addressing the visualisation of webpage hierarchies and resource dependencies in a mobile-friendly format. This gap represents the primary focus of my research.

Ethics Statement

This report is produced in good faith to encourage and facilitate colleagues to develop as professionals first and foremost.

No special approval was deemed necessary to be carried out when discussion of this project was held.

Aims

We aim to produce a library of datastructures and functions that can be composed together in order to ergonomically and optimistically produce a library of prepared messages for the LLM worker, that produce consistent and useful results.

This library is designed with principle of information transformation, analogous to mathematical studies of Lie Groups (Wikipedia contributors 2025a) in Category Theory (Wikipedia contributors 2025b).

Objectives

Our objective is to create and document a framework for prompt engineering that could be used as part of a note-taking process, in collaborative or solo setting, which aids the user by providing a toolbox of atoms of interaction with the LLM.

These atoms will be defined by tools and parsers that can be applied to any commercially available off-the-shelf LLM, in hopes of producing a useful toolbox for notetaking, creative writing, documentation writing, etc.

Keywords: large language models, notetaking, prompt engineering, collaborative editing

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